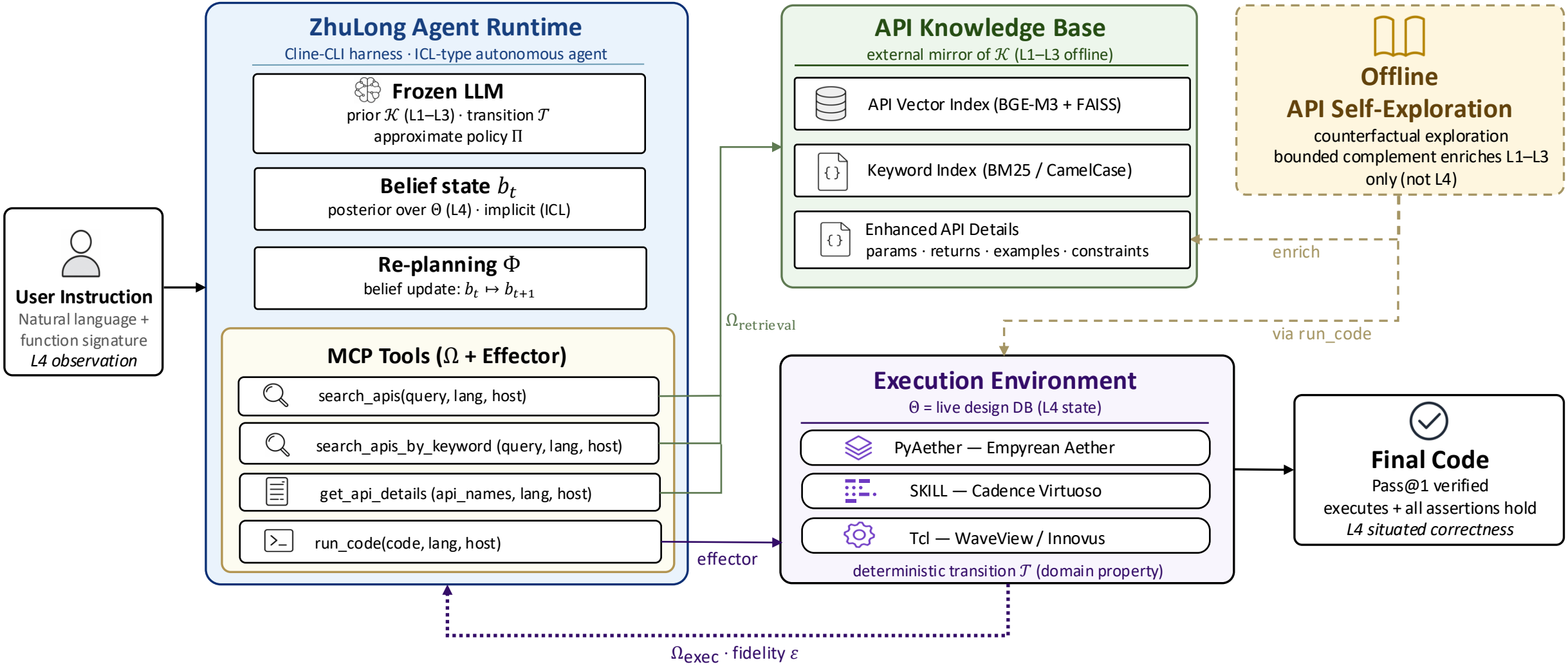

ZhuLong Agent Runtime
Cline-CLI harness · ICL-type autonomous agent
Frozen LLM
prior $\mathcal{K}$ (L1–L3) · transition $\mathcal{T}$
approximate policy Π
Belief state b_t
posterior over Θ (L4) · implicit (ICL)
Re-planning Φ
belief update: $b_t \mapsto b_{t+1}$
MCP Tools (Ω + Effector)
search_apis(query, lang, host)
search_apis_by_keyword (query, lang, host)
get_api_details (api_names, lang, host)
run_code(code, lang, host)
User Instruction
Natural language + function signature
L4 observation
API Knowledge Base
external mirror of $\mathcal{K}$ (L1–L3 offline)
API Vector Index (BGE-M3 + FAISS)
Keyword Index (BM25 / CamelCase)
Enhanced API Details
params · returns · examples · constraints
Offline API Self-Exploration
counterfactual exploration
bounded complement enriches L1–L3 only (not L4)
enrich
via run_code
$\Omega_{\text{retrieval}}$
Execution Environment
Θ = live design DB (L4 state)
PyAether — Empyrean Aether
SKILL — Cadence Virtuoso
Tcl — WaveView / Innovus
deterministic transition $\mathcal{T}$ (domain property)
effector
Final Code
Pass@1 verified
executes + all assertions hold
L4 situated correctness
Ω_{exec} · fidelity ε